

Tide retrospective: identifiability under analytic hypotheses

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

Seventh tide of the `germbij` arc, and the payoff of the previous one: the analytic instantiation¹ bridged finite analytic order to the chain’s factored growth hypothesis; this tide composes it with the merged 1D identifiability bound, so that the `germbij` Theorem 7.3 chain now runs from analyticity outright. The candidate was deliberated (as candidate B) in the previous tide and deferred by joint vote to “a short subsequent composition once A is stable”.

2 The thing formalised

One theorem, in `Laplace/AnalyticIdentifiability.lean`, sorry-free.² For $L_1, L_2 \geq 0$ analytic at 0 with differing germs (finite analytic order of the difference), dominated by $C_0 w^2$ on $[0, R]$, and a nonnegative bump $\psi \equiv 1$ on $[0, R]$: there exist m (the vanishing order of $L_2 - L_1$), $c > 0$ and $r_0 \in (0, R]$ such that for every t with $4 \leq r_0^2 t$, under the per- t integrability premises,

$$c \cdot t \cdot t^{-m-1/2} \leq \int (L_2 - L_1) \psi (e^{-tL_1} - e^{-tL_2}) dw.$$

The quantifier arrangement is the note’s: constants first, then all sufficiently large t . The hypotheses are now exactly the mathematical ones (analyticity, nonnegativity, quadratic domination, the bump) plus integrability; no growth or continuity hypothesis survives, both being discharged from analyticity inside the proof.

3 Proof strategy

Composition, with one derivation. The growth bound comes from the previous tide’s instantiation applied to $g = L_2 - L_1$ (analytic by `AnalyticAt.sub`). The merged 1D composite also wants continuity of the window integrand; this is derived rather than assumed: the eventual-analyticity lemma³ gives a ball of analyticity of radius ρ around 0 for both potentials, r_0 is shrunk to $\min(r_1, R, \rho/2)$, and on the admissible Laplace window (inside $[0, r_0]$ by the usual $2 \leq r_0 \sqrt{t}$ algebra) the integrand is continuous pointwise, closed by `fun_prop` from the two `ContinuousAt` facts. The final step is a single `exact` of the merged composite with the restricted hypotheses.

¹2026-08-09-02-55-tide-germbij-analytic.

²`analytic_pencil_difference_lower_bound`.

³`AnalyticAt.eventually_analyticAt`.

4 Roadblocks and resolutions

One iteration: the first attempt forgot that the 1D composite (unlike its multivariate sibling, which derives integrability on the window from the minorant) carries an explicit window-continuity hypothesis. The repair became the tide’s main content: discharging that hypothesis from analyticity, which is both stronger and closer to the note’s statement.

5 What was Mathlib, what was new

Mathlib: the `AnalyticAt` subtraction, eventual-analyticity, and continuity lemmas, the filter and metric idioms, and `fun_prop` for the compound continuity. New: the composition and the quantifier packaging.

6 Lessons

When composing against a merged theorem, read its full hypothesis list first (the 1D and multivariate composites deliberately differ in one hypothesis). And when a premise can be discharged from a structural hypothesis already present (here: continuity from analyticity), discharging beats propagating: the user-facing statement shrinks and the proof grows by only a dozen lines.

7 Follow-ups

- The multivariate analytic corollary: needs the leading-part instantiation in d variables (produce the scaled set S from a nonzero homogeneous leading part), the one genuinely new ingredient left.
- Turnkey integrability: derive the per- t premises from continuity plus compact support of ψ , removing the last non-mathematical hypotheses.
- Note annotation: tag Theorem 7.3 with the analytic corollary once merged (it certifies the analytic-hypothesis form directly).